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MANAGING CITIZENS' EXPECTATIONS OF PUBLIC SERVICE PERFORMANCE: EVIDENCE FROM OBSERVATION AND EXPERIMENTATION IN LOCAL GOVERNMENT

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Citizens' expectations of public service performance influence their attitudes and behaviour towards services, including satisfaction, choice of service and political voice about them. However, there has been little research on what sets expectations. This paper assesses the effects of prior service performance and information about prior performance on two forms of citizen expectations, positive expectations of what performance will be and normative expectations of what performance should be. In an observational study, prior performance is positively related to expectations of what performance will be. Prior performance is positively related to high normative expectations but is unrelated to low normative expectations. The resilience of normative expectations suggests that poor performance will trigger dissatisfaction and citizen response rather than lowering expectations creating passive acceptance. In a field experiment, performance information effects are found for positive but not normative expectations. Providing information about excellent performance raises positive expectations and providing information about poor performance lowers positive expectations; negativity bias is evident with information about poor performance having a larger effect. Performance information that is credible to citizens can be used to manage citizens' positive expectations but their normative expectations are less amenable to influence by this route.

Consumers' expectations about private goods and services have long been discussed but recently there has been a growth of interest in citizens' expectations about public services. Citizens' expectations of public service performance influence their satisfaction with services (Roch and Poister 2006; Van Ryzin 2004, 2006; James 2009; Poister and Thomas 2011), their choice of services and their political 'voice', including voting for those in charge of services and lobbying for improvement (Hirschman 1970; Lyons and Lowery 1989; Lyons *et al*. 1992; Dowding and John 1996; James and John 2007; Dowding and John 2008; Boyne *et al*. 2009). However, there has been little research directly examining the factors that influence citizens' expectations of public service performance.

This article analyses the relationship between the prior performance of services and expectations and the relationship between provision of information about prior performance and expectations. Prior performance has been shown to have a strong positive relationship with the subsequent performance of public services (Meier and O'Toole 2002; Boyne *et al*. 2009). If prior performance influences citizens' expectations about this performance then the expectations are grounded in prior performance. However, if not, citizens' expectations are disconnected from service performance making it difficult for public managers and politicians to engage citizens in meaningful debate, as has been found for many areas of politics (Downs 1957; Lupia and McCubbins 1998). Without expectations that reflect performance it is also difficult for citizens to satisfy their preferences by exercising the choice over public services that is a feature of many public service systems. However, an increasing amount of information about service performance is now publicly available and the provision of this information has been shown to affect

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citizens' perceptions of performance and their satisfaction with services (James 2011). This article analyses the effects of providing performance information on citizens' expectations. If the provision of information influences expectations this effect suggests a relatively low cost way to inform, or even manage, citizens' expectations through the provision of information.

The first section of the article defines positive and normative concepts of expectations about public service performance and sets out the theory of the relationship between expectations and prior performance and between the provision of performance information and expectations. The second section describes an observational study and field experiments to evaluate hypotheses derived from the theory in the context of the overall set of services provided by local governments in England. The use of field experiments in public administration is unusual despite benefits for obtaining unbiased estimates of causal effects (Shadish *et al*. 2002; Green and Gerber 2003). The third section describes the findings and the fourth section draws conclusions and suggests directions for future research.

EXPECTATIONS ABOUT PUBLIC SERVICES

The literature on consumer expectations about private sector goods and services has been criticized for failing to offer clear and consistently used definitions (Teas 1993, 1994) so it is important to avoid such problems in the context of public services. Expectations are future oriented subjective beliefs and attitudes and concern a wide range of objects. Following James (2009, p. 109), this article defines expectations as 'judgements of what individuals or groups think either will or should happen under particular circumstances'. In this sense, positive expectations are what citizens think service performance will be and normative expectations are what citizens think performance should be. Normative expectations do not entail that citizens think the desired outcomes will necessarily occur and positive expectations do not entail that they think what will occur is necessarily desirable.

Positive expectations of objects are widely used in economics for examining individuals' decision making under partial information. Economic modelling commonly assumes that people from probabilistic expectations for unknown quantities, usually as part of judgement and decision-making processes that attempt to maximize expected utility. When expectations are combined with knowledge of preferences researchers are able to make predictions about behaviour. Economists have traditionally made assumptions about expectations that are not directly empirically assessed and have used observed choice behaviour to uncover preferences. However, the assumptions are not tenable in many contexts and economists have started to conduct surveys of expectations to gain information about subjective probabilities of particular events of interest (Manski 2004). Positive expectations also feature in many models of political participation. In studies of the relationship between the economy and voting, the expected value of outcomes has been found to be important for vote choice (Lewis-Beck and Paldam 2000).

There has been little systematic effort to address positive expectations about the performance of public services until recently with a series of studies beginning to apply these measures (Van Ryzin 2004, 2006; James 2009). Some policy related work has also been undertaken by public services providers and opinion pollsters with probably the largest investment being in a rolling cross sectional survey of citizens' expectations of whether services in health, education and local services will get better or worse as part of a 'delivery index' for UK public services (MORI 2009).

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Normative expectations of what citizens think performance should be imply that they have a view of what is reasonable or desirable. These expectations embody a preference for a particular state of affairs either in an ideal world or in another specific background context. In the literature on private sector services the benchmark is often that of what an 'excellent' company should do, for example, as used by the widely applied SERVQUAL model of satisfaction with consumer goods and services (Parasuraman *et al*. 1985, 1988). This benchmark sets desirable features of a company or service in a particular context. For public services, there has been little research on normative expectations. The context in which judgements about public services are made includes the taxes paid, other resources available, and conditions influencing the difficulty of service provision. Respondents have been asked what a 'reasonable' level of service would be for health care services (Appleby and Alvarez-Rosete 2003). Measures have been developed for citizens' expectations of what ought to be the performance level for overall services provided by local public service organizations (James 2009) and these measures have been adapted and extended for specific services including household waste collection services (James 2009) and state highway services (Poister and Thomas 2011).

Normative and positive expectations have different roles in models of how expectations influence satisfaction with public services. Satisfaction is increasingly used as a measure of service performance to supplement managerial measures (Miller and Miller 1991; Bouckaert *et al*. 2005; Roch and Poister 2006; James 2009; Van de Walle and Van Ryzin 2011). For example, satisfaction surveys in England have in recent years been a requirement for local authorities and are used in a range of oversight activities (Office of Public Service Reform/MORI 2002; ODPM 2004). Models of expectation 'confirmation/disconfirmation' suggest that satisfaction is positively related to performance minus expectations such that higher expectations will be associated with lower satisfaction. This model has been examined for private services, and increasingly in for public services (Van Ryzin 2004; Roch and Poister 2006; Van Ryzin 2006; James 2009). Van Ryzin (2004, 2006) found mixed results for the relationship between positive expectations and satisfaction for local public services in the US. However, he found consistent support for a subtractive model of expectations, with experienced quality minus expected quality being positively related to satisfaction (Van Ryzin 2006). Turning to normative expectations, this form of expectation has been found to have a clear negative relationship with satisfaction for overall services provided by local government. The variable of perceived performance minus expected normative performance has also been found to be positively related to satisfaction (James 2009). The predicted probability of citizens being satisfied with overall services provided by their local government was just 0.2 for citizens who were disappointed through performance failing to meet their normative expectations but rose to almost 0.8 where these expectations were met (James 2009, p. 114). Evidence about the specific services of household waste services (James 2009, p. 115) and state highway services (Poister and Thomas 2011) show the same direction of relationship. These findings suggest that satisfaction with public services can be increased by lowering normative expectations about services as well as by increasing the performance of those services.

Positive and normative expectations don't just influence attitudes but also citizens' behaviour towards services. If citizen satisfaction can be improved by managing normative expectations it is less likely that citizens will vote against those in charge of public services or complain about services (Dowding and John 1996; James and John 2007; Dowding and John 2008; Boyne *et al*. 2009). Positive expectations are also negatively related to complaints about services and voting against incumbents because they relate to the

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performance citizens think they will get if they use a service. Positive expectations are positively related to service use for the same reason, and citizens choose services that they have high positive expectations about relative to other service options where they have a choice of provider available to them (Dowding and John 1996, 2008).

There has been little work on the factors that influence citizens' expectations of public service performance. Many different influences have been identified for the expectations of consumers of private sector services (Zeithaml *et al*. 1993). The dominant approach to the process of expectations formation in economics is that decision makers have 'rational' expectations (Sargent 2008). Rational expectations suggests that citizens have a model of how public services operate, for example a production function for services, and use this model to forecast current and future performance. On average, individuals are correct even if individual forecasts are prone to error. However, expectations have been found not to operate in this way in many contexts similar to those relating to public services, for example for future income, life expectancy or returns to schooling. Alternative models, especially of adaptive expectations in which expectations are influenced by the previous values of the variable of interest, have been found to be more accurate (Manski 2004). In an adaptive model of expectations of service performance citizens use prior performance to inform their positive expectations of performance, suggesting the following hypothesis for local government services:

> H1 Prior performance of local government services is positively related to citizens' positive expectations of performance.

Citizens are known to respond to extreme performance levels more than differences in performance across the mid range of performance, for example, in punishing incumbent politicians electorally only for poor performance (James and John 2007; Boyne *et al*. 2009; James 2011). On this basis, a second version of the prior performance hypothesis is suggested:

> H2 Extremes of prior performance of local government services are related to citizens' positive expectations of performance; excellent prior performance raises positive expectations and poor prior performance lowers positive expectations.

Normative expectations are also informed by prior performance although the relationship is more complex than that for positive expectations. Citizens' conceptions of what performance ought to be imply some valuation of performance levels. These conceptions are similar in this respect to individuals' fundamental valuations of goods and services defined as their assessments of the benefits from these goods and services. Individuals' estimates of fundamental values have been shown in many contexts to be influenced by anchors set by historical experience that have a big influence on the valuations (Ariely *et al*. 2003). For example, historical prices given to individuals often set the prices these individuals value objects at, rather than individuals having exogenous preferences which determine their assessments of what the price should be. These findings suggest that citizens' normative expectations of performance are anchored by the actual prior performance of the local unit that the citizens have previously experienced. This hypothesized relationship is assessed in two forms, with the latter form expressing the previously discussed tendency for extremes of prior performance to have the biggest impact:

> H3 Prior performance of local government services is positively related to citizens' normative expectations of performance.

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H4 Extremes of prior performance of local government services are related to citizens' normative expectations of performance; excellent prior performance raises normative expectations and poor prior performance lowers normative expectations.

This paper also examines the effect of providing information about the prior performance of public services. Performance information is part of the transmission mechanism by which prior performance influences expectations and can have its own independent effects. Citizens often don't know much about government policies and performance (Delli Carpini and Keeter 1996; Lupia and McCubbins 1998). Information about public services comes from many sources, especially personal experience, word of mouth, the media, public auditors and public service providers themselves (Heinrich 2003; Moynihan 2008). Citizens' lack of information is particularly acute for services that they do not themselves directly receive, or which they interact with selectively or only from time to time. Providing information to citizens has been shown to have effects in other contexts. Information about excellent performance raises citizens' assessments of performance and information about poor performance lowers their assessments. Citizens' satisfaction with services is raised by provision of information about excellent performance and lowered by information about poor performance (James 2010).

If information about performance influences expectations then public managers and politicians may be able to use the provision of information to manage these expectations. The expectations of consumers of private sector services are influenced by company communications to customers (Parasuraman *et al*. 1985) but there has been no research about whether this is possible for public services. The hypotheses relate to the provision of information about extremes of performance using information provided by public audit bodies that are formally independent of service providers themselves because these levels and this source have been found to be most influential on citizens' perceptions and satisfaction (James 2011):

H5 Providing information about excellent prior performance raises positive expectations and providing information about poor prior performance lowers positive expectations.

H6 Providing information about excellent prior performance raises normative expectations and providing information about poor prior performance lowers normative expectations.

RESEARCH METHOD

The first part of this research used an observational study of the performance of local governments as assessed by auditors and a survey of citizens in England to analyse the relationship between expectations and prior performance. The expectations were about the overall performance of a set of local public services provided by the citizens' local government, consisting of environmental and waste services, social and educational services, and a set of sporting and leisure services. In England, there is a defined set of core services that has to be provided by each local government across the country and which dominate the activities of these bodies. Expectations about the performance of the overall set of services are important because they inform the citizens' voting for local politicians seeking to control these bodies, and facilitate choice by informing this judgement (James and John 2007; Dowding and John 1996, 2008).

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The survey participants were citizens who had lived in their local area for at least a year and so would have come into contact with services over that period. The participants were a randomly selected group of 9,500 citizens from a representative bank of 100,000 internet users run by the survey firm YouGov. The survey group had similar socio-economic, ethnic and gender characteristics to the general English population. The survey was conducted in summer 2005 and the response rate to questions about expectations was 26.3 per cent. Internet panels are receiving increasing acceptance in the social sciences; the 2005 British Election Survey found little or no difference between results gained from an internet poll and conventional random probability door to door survey (Sanders *et al*. 2006).

Positive expectations were measured on a five point scale asking citizens what they would expect if they were to interact with their local government's services. The approach was to relate expectations of performance to a benchmark of excellent performance as conceived by each respondent. Respondents were also surveyed about what excellence should consist of for specific features of services to establish that a common view of excellence existed from which individual expectations might depart in terms of the proportion of time services meet this standard. Each response option for expectations had a specific text description. The highest response point was 'performance will be excellent all of the time', there were three descending points described as 'will be excellent most of the time', 'will be excellent some of the time' and 'will rarely be excellent'. The lowest point was described as 'will never be excellent'. Normative expectations of what the current performance of local public services should be were also solicited on a five point scale. The highest point was described as 'should be excellent all the time', there were three descending points described as 'should be excellent most of the time', 'should be excellent some of the time' and 'should rarely be excellent'. The lowest point was described as 'should never be excellent'.

Prior performance was measured using data from the Comprehensive Performance Assessment of local governments for 2004. This information was checked by independent auditors of local government with the data obtained from the Audit Commission. What counts as 'performance' is relative to stakeholder concerns and a range of judgements about its measurement and evaluation. The system for England has been criticized for not measuring all aspects of service performance with full reliability. However, the system is one of the most developed performance measurement regimes ever instituted and citizens' attitudes and behaviour have been found to respond to performance measured in this way, suggesting that auditors and citizens share at least some understanding of performance (James and John 2007; Boyne *et al*. 2009; James 2011). Two measures of performance were included:

(1) A measure across the full range of performance;
(2) A measure identifying excellent and poor local government service performers with excellent performers defined as those rated 'excellent' and the poor performers defined as those rated 'poor' or 'weak' by the Comprehensive Performance Assessment.

A range of factors other than prior performance have previously been identified as influential on expectations so several control variables were incorporated in the analysis. Ethnicity, defined as non-white British, and age were included because both have been suggested to be positively related to positive and negatively related to normative

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expectations. Education level, measured as years of education after the age of 15, was included because increasing education has been suggested as being positively associated with normative expectations and negatively associated with positive expectations. Household income was incorporated because it has been suggested as positively related to normative expectations and gender was included because women have been suggested as having lower normative expectations (Cabinet Office 2000; Duffy 2000; Performance and Innovation Unit 2001; Office of Public Services Reform/MORI 2002). Although the evidence base for these relationships, especially about their direction and strength, is limited there is enough evidence of some form of relationship to justify their inclusion in the analysis. The values for these variables were gathered from respondents in the survey. A further control variable included was Council Tax for the local governments for 2005/6 using data from the Department of Communities and Local Government because higher levels of resources are generally acknowledged as beneficial for performance and could raise both forms of citizens' expectations. Finally, a measure of the socio-economic context in which services are provided was included because this context is often seen as influencing the challenge presented to service providers, with poorer areas making provision more difficult. More challenging contexts lower expectations of what performance will be and potentially what it should be. To measure this context, the Index of Multiple Deprivation for each area in 2004, the most recent year available, was adopted using data from the Department for Communities and Local Government.

The second part of the study analysed the effect of providing performance information on citizens' positive and normative expectations using field experiments with randomization of this intervention. Experiments involve the manipulation of hypothesized causal factors to examine their effects (Shadish *et al*. 2002, pp. 1–12). Randomized experiments involve the alternative interventions, for example a simple dichotomy between intervention and control, being allocated to experimental units, such as people or organizations, by chance. Randomization creates groups that are, on average, similar and differences in outcomes between groups once treatments are applied can be used to estimate the causal effect of the intervention. Systematic correlation between the intervention condition and other known or unknown characteristics of participants is avoided; randomization avoids selection bias where those subject to intervention have particular characteristics that also themselves affect the outcome.

The use of experiments with random allocation rather than observational techniques is valuable in the context of information and expectations because the causal relationship is difficult to disentangle. For example, citizens in a well performing area who have high positive or normative expectations are probably more engaged in local services and are more likely to seek out information about the performance of services than are citizens in the same area who have lower expectations. These relationships might lead an observational study to overestimate the positive effect of providing information on citizens' expectations. Random allocation of the information intervention overcomes this selection bias by making the groups of citizens who do and do not receive performance information on average equal in their initial expectations.

The experiments were conducted in a field setting (Green and Gerber 2003; Harrison and List 2004). The field context was a naturalistic setting in the local government areas using a subject pool of local citizens. Two experiments were conducted. One experiment was in a local government area with provision of performance information about excellent performance and the other experiment was in a local government area with provision of information about poor performance. The information came from the Comprehensive Performance

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TABLE 1 *Characteristics of participants in the field experiments*

Characteristic	Excellent performance experiment	Poor performance experiment	England and Wales
Age in years, mean	41.8	39.8	38.7
Gender, female (%)	53.7	49.7	51.3
Non-white (%)	1	6	9.1
Unemployed (%)	2	2.1	5.0

Source: 2001 Census (Office of National Statistics); surveys conducted as part of this research.

Assessment of each area. The field context provides good external validity of findings to citizens in contexts where elected governments working with professional managers are responsible for a set of local public services. The characteristics of participants in the experiments and those of the general population of England and Wales are summarized in table 1.

Participants were recruited from passers-by in the street within urban communities in each local government area on different days over a two-week period in July 2010. All participants were told that the research was being conducted by a University and that their participation was voluntary. The estimated required sample sizes were calculated using a guide to expected differences in means for expectations outcomes provided by earlier research on the means and standard deviations of perceived performance variables for two groups of citizens' where one group received performance information and one group did not. Information about good performance had a Cohen's d effect size of 0.455 in raising citizens' satisfaction with local public services (James 2010). The calculation used the sampsi command in STATA with power set at 0.9 and alpha 0.05, suggesting a sample size of 97 in each group. In the end, 88 participants were recruited to each group in the experiment providing information about excellent performance and 89 and 90 participants were recruited to the groups in the experiment providing information about poor performance. These sample sizes were close to those suggested as necessary to find an effect if one of the estimated magnitude, based on the prior research, was present.

Having agreed to participate, participants were randomized to either intervention with associated information provision in a document also containing a survey or to control documentation containing the same survey. The randomization sequence was established by a market research organization in their central office and those conducting the survey in the field were not able to choose citizens' allocation to each group and were blind to which group participants were entering because all the documents looked the same externally. The first section of the document contained a survey with questions to gather data for the variables age, gender, employment and ethnicity. The document further contained information about local services provided by the council and, for the treatment groups only, information about either excellent (in one local government area) or poor prior local service performance (in the other area). Participants were told the information came from independent auditors of local government. In all other respects the documents for those given information and those not given information were identical. The final section of the document contained questions about positive and normative expectations using the same measures of expectations as used in the observational study.

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TABLE 2 *Regression models of prior performance and positive expectations of performance*

	Model 1 (linear)	Model 2 (linear)	Model 3 (ordered probit)	Model 4 (ordered probit)
Age (years)	0.002 (0.001)*	0.002 (0.001)*	0.003 (0.002)	0.003 (0.002)
Gender (female)	0.084 (0.031)***	0.085 (0.031)***	0.120 (0.045)***	0.121 (0.045)***
Years of education after the age of 15	−0.022 (0.012)*	−0.024 (0.012)*	−0.032 (0.018)*	−0.035 (0.018)**
Ethnicity (non white British)	−0.087 (0.049)*	−0.089 (0.048)*	−0.126 (0.070)*	−0.129 (0.070)*
Council tax rate 2005	−0.001 (0.000)****	−0.001 (0.000)****	−0.001 (0.000)****	−0.001 (0.000)****
Annual household income	0.000 (0.009)	0.001 (0.009)	0.001 (0.013)	0.001 (0.013)
Level of deprivation	−0.003 (0.001)**	−0.003 (0.001)**	−0.005 (0.002)**	−0.004 (0.002)**
Performance assessment score	0.037 (0.018)**		0.053 (0.026)**	
Excellent performers		0.116 (0.070)*		0.170 (0.103)*
Poor performers		−0.197 (0.062)***		−0.282 (0.086)***
F	(8,345) 6.98****	(9,345) 7.88****		
R squared	0.021	0.024		
Wald chi squared			(8) 54.97****	(9) 70.15****
Pseudo R squared			0.009	0.010
N	2495	2495	2495	2495

Note: Robust standard errors (adjusted for 346 clusters in local governments).
*indicates $p < 0.1$; **indicates $p < 0.05$; ***indicates $p < 0.01$; ****indicates $p < 0.001$. In tests of the parallel regression assumption in the ordered probit models using the approximate likelihood ratio test of equality of coefficients across response categories the null was not rejected. For model 3, chi squared (24) = 25.45 ($p = 0.381$); for model 4, chi squared (27) = 27.10 ($p = 0.458$).

RESEARCH FINDINGS

The hypotheses about the relationship between prior performance and expectations from the observational study were assessed using multivariate statistical analysis. The summary statistics for the models in are reported in table A1 in the appendix. The hypothesized positive relationship between prior performance and positive expectations of performance (H1) is supported by results from analysis using linear regression. Model 1, reported in table 2, has a small positive coefficient of 0.037 for the prior performance variable, having the highest level of prior performance is associated with 0.148 higher expectations compared to having the lowest level of prior performance (on an expectations scale ranging from 1 to 5). Hypothesis H2, that extremes of performance are related to positive expectations, is supported and the relationship is stronger for extreme levels of performance than across the middle range of performance scores, as shown in model 2. Being in the excellent prior performance category is associated with 0.116 higher expected performance; being in the poor prior performance category is associated with 0.197 lower expected performance. These findings show evidence of negativity bias because of the larger absolute value of the coefficient for poor prior performance compared to that for excellent prior performance.

A more conservative modelling approach treats the expectations variables as ordered categories rather than interval level data and uses the ordered probit models 3 and 4 in table 2. The results of this analysis support the findings from the linear models. Prior performance is positively related to positive expectations; excellent prior performance is associated with higher positive expectations and poor prior performance is associated with lower positive expectations. Ordered probit model coefficients are not amenable to straightforward direct interpretation but the models allow the calculation of predicted

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TABLE 3 *Predicted probabilities of levels of positive expected performance for a local government with excellent prior performance*

Expected performance	Middle range of prior performance	Excellent prior performance	Difference (with 95 confidence interval, lower bound, upper bound)
Never excellent	0.010	0.006	−0.004 (−0.008, 0.000)
Rarely excellent	0.120	0.092	−0.029 (−0.035, −0.023)
About half of the time	0.484	0.450	−0.034 (−0.035, −0.033)
Mostly excellent	0.355	0.408	0.053 (0.050, 0.055)
Excellent all the time	0.031	0.045	0.014 (0.000, 0.032)

probabilities for citizens having particular levels of expected performance associated with different values of the independent variables. Citizens in local governments with excellent prior performance have lower predicted probabilities of reporting the lowest two levels of expected performance and higher predicted probabilities of reporting the two highest levels of expected performance as shown in table 3. However, the differences are not large; the biggest difference in predicted probability is an increase of 0.053 for those reporting expected performance as mostly excellent.

Citizens in local governments with poor prior performance have lower predicted probabilities of reporting the two highest levels of expected performance and higher predicted probabilities of reporting the two lowest levels compared to citizens in the middle range of performers, as shown in table 4. The largest difference is a decrease of 0.088 for reporting mostly excellent expected performance. The differences for poor prior performance are larger in absolute terms than is the case of excellent prior performance, suggesting negativity bias in the relationships.

Turning to citizens' normative expectations about performance and hypotheses H3 for prior performance and H4 for extremes of prior performance, the relationships are only partially as hypothesized. The ordered probit models with the same explanatory variables as models 3 and 4 violate the parallel regression assumption that the coefficients are the same across response categories, which suggests that the restricted functional form positing the same relationship across all levels of normative expectations is not appropriate. In models of normative expectations with same independent variables as models 3 and 4, the approximate likelihood ratio test of equality of coefficients was rejected for the analogue to model 3 chi squared (24) = 65.48 (p = 0.000); and for the analogue to model 4, chi squared (27) = 67.66 (p = 0.000). This finding also implies that linear models are not appropriate because they have an even more restrictive functional

TABLE 4 *Predicted probabilities of positive expected performance levels for prior poorly performing local government*

Expected performance	Middle range of prior performance	Poor prior performance	Difference (with 95 confidence interval, lower bound, upper bound)
Never excellent	0.010	0.020	0.010 (0.002, 0.018)
Rarely excellent	0.118	0.176	0.058 (0.049, 0.068)
About half of the time	0.482	0.516	0.035 (0.032, 0.037)
Mostly excellent	0.360	0.273	−0.088 (−0.090, −0.085)
Excellent all the time	0.031	0.016	−0.015 (−0.023, −0.008)

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TABLE 5 *Probit regression models of prior performance and normative expectations of performance*

	Model 5: High expectations	**Model 6: High expectations**	**Model 7: Low expectations**	**Model 8: Low expectations**
Age (years)	−0.003 (0.002)	−0.003 (0.002)	−0.011 (0.003)	−0.011 (0.003)
Gender (female)	0.179 (0.057)***	0.175 (0.057)***	−0.078 (0.062)	−0.079 (0.061)
Years of education after the age of 15	−0.061 (0.020)***	−0.064 (0.020)***	−0.008 (0.029)	−0.006 (0.030)
Ethnicity (non white British)	−0.004 (0.086)	−0.006 (0.086)	0.344 (0.106)**	0.343 (0.107)***
Council tax rate 2005	0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)
Annual household income	−0.008 (0.014)	−0.009 (0.014)	−0.040 (0.019)**	−0.040 (0.019)
Level of deprivation	0.000 (0.002)	0.000 (0.002)	0.000 (0.003)	−0.000 (0.003)
Performance assessment score	0.069 (0.026)***		−0.034 (0.037)	
Excellent performers		0.251 (0.083)***		−0.271 (0.171)
Poor performers		−0.039 (0.096)		0.156 (0.160)
Wald chi squared	(8) 35.73****	(9) 40.83****	(8) 33.30****	(9) 37.66****
Pseudo R squared	0.010	0.009	0.021	0.022
N	2499	2499	2499	2499

Note: Robust standard errors (adjusted for 346 clusters in local governments).
*indicates $p < 0.1$; **indicates $p < 0.05$; ***indicates $p < 0.01$; ****indicates $p < 0.001$.

form. The approach taken is to model the highest level of normative expectations and the lowest level of normative expectations as dependent variables in separate models to enable comparisons with the rest of the range of expectations for each model.

The findings about prior performance and normative expectations are only partially consistent with the hypothesized relationships, as shown in table 5. Model 5 shows a positive relationship between prior performance and high normative expectations of performance and model 6 shows a positive relationship between excellent prior performance and high normative expectations, providing support for H3 and H4 respectively. Using the data from model 6 to calculate predicted probabilities, prior high performance raises the predicted probability of reporting high normative expectations from 0.410 to 0.509, an increase of 0.099 (with the 95 per cent confidence interval for this difference having an lower bound of 0.035 and an upper bound of 0.164). However, contrary to H4, poor prior performance is not associated with a lower predicted probability of citizens having high expectations compared to the rest of the range of performance. The findings are asymmetric, with excellent prior performance raising the predicted probability of high normative expectations but poor prior performance not lowering the predicted probability of reporting this level of expectations. The relationships between prior performance and the lowest level of normative expectations, and the extremes of excellent and poor prior performance and the lowest level of normative expectations are not as hypothesized in H3 and H4 respectively.

The findings for hypothesis H5 about the effects of providing performance information on positive expectations using evidence from the field experiments are summarized in table 6. The analysis was conducted on an intention to treat basis and there was no drop out of participants who received either the information or control documentation in the experiment; data was available for all those initially randomized into the experimental groups. The group of citizens who received information about excellent performance had a mean positive expected performance score of 3.716 (standard deviation = 0.606) and those without had a mean score of 3.500 (standard deviation = 0.587). There is strong

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TABLE 6 *Positive and normative expectations and field experiments providing information about excellent performance or poor performance of local government*

Dependent variables	Statistical analysis	Information about government with excellent performance	Information about government with poor performance
Positive expectation about performance	One tailed t-test (df) statistic	t(174) = 2.040***	t(154) = −5.679****
	Wilcoxon rank sum/ Mann-Whitney test statistic	z = 2.109** n = 176	z = −4.730**** n = 179
Normative expectation about performance	One tailed t-test (df) statistic	t(162) = 1.184	t(176) = −0.142
	Wilcoxon rank sum/ Mann-Whitney test statistic	z = 0.736 n = 176	z = 0.233 n = 179

*indicates $p < 0.1$; **indicates $p < 0.05$; ***indicates $p < 0.01$; ****indicates $p < 0.001$. The variance ratio test of the equality of variance assumption of t test with a null of ratio = 1 was undertaken in each case: For experiment about good performance and positive expectations (f = 0.941, df = 87, 87, p = 0.776) and normative expectations (f = 1.717, df = 87, 87, p = 0.013 a t test with an assumption of unequal variance was undertaken in this latter case. For experiment about bad performance and positive expectations (f = 0.456, df = 89, 88, p = 0.000) and normative expectations (f = 1.389, df = 89, 88, p = 0.126) so a t test assuming unequal variance was undertaken in the former case.

support for this part of H5 with the difference between these scores being statistically significant at a conventionally strong level ($p < 0.01$) in a t test of comparison of means. An indication of the substantive importance of the difference in means is given by the Cohen's d effect size measure. This measure is calculated here by dividing the difference in means by the pooled standard deviation of the two groups. On this basis, providing information about the government with excellent performance raises positive expectations of performance with a Cohen's d effect size = 0.356. A more conservative approach assuming responses on the scale for measuring expectations only as ordered categories was also conducted to check the robustness of the findings across this variation in assumption. The rank sum test statistic showed a statistically significant difference at the $p < 0.05$ level.

Turning to the hypothesized effect of information about poor local government performance on positive expectations, as the second aspect of H5, the group of citizens who received information about poor performance had a mean positive expected performance score of 2.337 (standard deviation = 1.389) which was lower than those without this information, who had a mean score of 3.344 (standard deviation = 0.938). This part of H5 is supported with the difference between these scores being statistically significant at a conventionally very strong level ($p < 0.001$) in a t test of comparison of group means. An indication of the substantive importance of the difference in means is given by the Cohen's d effect size = −0.784. A more conservative approach assuming responses on the scale for measuring expectations only as ordered categories using a rank sum test statistic showed a statistically significant difference at the $p < 0.001$ level.

The hypothesized effect of information about excellent local government performance raising normative expectations, the first part of H6, was not supported by the data. Whilst the mean score for those in the group receiving information about excellent performance was higher at 4.386 (standard deviation = 0.596) than for those in the group that did not, who had a mean of 4.261 (standard deviation = 0.780), the associated statistical test failed to reach the $p < 0.1$ level of significance. The associated Cohen's

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d effect size is 0.180. The more conservative approach using expectations measured as ordered categories was associated with a rank sum statistical test also failed to meet the $p < 0.1$ level of significance. Information about poor performance does not lower normative expectations either and the second part of H6 was not supported. The mean score for those in the group receiving information about bad performance was 3.943 (standard deviation = 0.902) which was very slightly higher than those who did not who had a mean of 3.922 (standard deviation = 1.062). The Cohen's d effect size measure is only −0.021.

The findings reveal negativity bias in the effect of performance information on citizens' positive expectations. The absolute value of the effect size for information about poor performance is just over twice the size of that for information about excellent performance. However, the citizens in the areas in which the experiments were conducted differ on a range of observed and unobserved factors between the areas. These differences do not influence the internal validity of the experiments within each area but they do sound a note of caution in comparing results across the areas. Despite this issue, the measured differences between the areas show that they are broadly similar socio-economically and demographically suggesting a reasonable degree of confidence in the finding of negativity bias.

CONCLUSION

Citizens' positive expectations about what local government public service performance will be are, unsurprisingly, positively related to the prior performance of local government. However, extremes of excellent or poor performance have a stronger association with positive expectations than differences within the mid range of performance. Performance information effects on citizens are also evident. The field experiments show that the provision of information about excellent performance raises positive expectations of what performance will be and information about poor performance lowers these expectations. The findings extend the scope of evidence about performance information effects on citizens that have previously been shown for citizens' perceptions of current performance and their satisfaction with public services (James 2011). Performance information provides citizens with information at low cost to overcome their rational ignorance and facilitates their participation in the democratic control of services, for example in local voting or public debates about services. The information is valuable especially about services they don't directly or regularly use themselves where there is little incentive to try and become informed by other routes. Even for services citizens do themselves use, information is valuable because it helps them exercise choice as users of services through knowledge of what they expect to receive.

Negativity bias is evident in both prior performance and performance information effects on positive expectations. These findings are consistent with a growing body of research that suggests that negativity bias is widespread in citizen and societal responses to public service performance. Negativity bias has previously been found in electoral responses to local government performance with voters punishing performance below the mid range but not rewarding performance above this level (James and John 2007; Boyne *et al*. 2009). Negativity bias has also been found in media reporting of performance (James 2004; Hood and Dixon 2010) and dissatisfaction rather than satisfaction is recognized as driving much citizen action about public services. Concern with handling blame is increasingly recognized as a major influence on how public

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service systems are designed and operate (Lyons *et al*. 1992; Hood 2002; James 2004; Hood 2011).

The relationship between prior performance and normative expectations of what service performance should be is much less clear than the relationship between prior performance and positive expectations. Prior performance across the full range of scores, and excellent prior performance, are both positively related to having high normative expectations of service performance. However, poor prior performance is not related to being less likely to have high normative expectations and neither excellent nor poor prior performance is related to having low normative expectations. Higher normative expectations are harder for service performance to meet than lower expectations. Expectations disconfirmation models in which satisfaction is positively related to perceived performance minus normative expectations show that citizens who are disappointed by service performance are less satisfied (James 2009; Poister and Thomas 2011). The findings of this study suggest that if performance fluctuates over time, prior excellent performance will increase the proportion of citizens with high normative expectations but poor prior performance will not lower this proportion. This process makes citizens progressively more difficult to satisfy over time. Future research should consider the relationship between prior performance, performance information and expectations over time, in a similar way to work conducted on expectations in the private sector context (Boulding *et al*. 1993).

The resilience of normative expectations to poor prior performance and information about performance reinforces the message of earlier research showing that poorly performing public services leads to dissatisfaction and political protest including voting against political incumbents presiding over poor performance (Lyons *et al*. 1992; James and John 2007; Boyne *et al*. 2010). The resilience of normative expectations suggests that it would be difficult for politicians or public managers to use performance information to manage these expectations. It is difficult to uncover the influences on attitudes towards public services that entail complex evaluative judgements (Van de Walle and Bouckaert 2003; Bouckaert *et al*. 2005). However, future research should explore the extent to which normative expectations are influenced by information about other aspects of services, the resources used in their production, and production processes. The way discussions are structured is also important as is recognized by a large literature on the psychology of judgement and decision making.

The field experiments show that performance information influences citizens' positive expectations. However, the effects may not be exactly the same if local public managers or politicians themselves presented the information as coming from their own assessments in an attempt to manage these expectations. The information in the experiments was given as being from independent public auditors which was a credible source to citizens who may be less inclined to trust information provided by politicians or service providers themselves because of the incentives that these groups have to misrepresent performance. These incentives also suggest that local public managers and politicians will not always be inclined to provide performance information provided by public auditors to their citizens. Those in charge of well performing organizations might benefit from releasing performance information to raise positive expectations because it makes citizens more positively disposed towards those formally responsible. Benefits for those in charge of poorly performing organizations are less clear because information about poor performance lowers positive expectations, perhaps damaging the reputation of the organization, but does not lower normative expectations which might otherwise make citizens easier to satisfy. This situation justifies higher tier 'regulatory' public authorities compelling the

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release of validated performance information by local bodies or providing it themselves directly to citizens (Hood *et al*. 1999; James 2000; Hood and Dixon 2010; James and Wilson 2010).

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APPENDIX

TABLE A1 *Summary statistics for variables in regression models*

	Mean	Standard deviation	Minimum	Maximum
Positive expectations	3.280	0.758	1	5
Normative expectations	4.268	0.731	1	5
Age (years)	42.478	14.094	19	76
Gender (female)	0.487	0.499	0	1
Years education after age 15	3.266	1.401	1	8
Ethnicity (non white British)	0.099	0.299	0	1
Council tax 2005	1215.095	95.258	614	1428
Annual household income	3.670	1.923	1	8
Deprivation	25.232	12.124	4.43	61.34
Performance assessment score	3.934	0.907	1	5
Excellent performers	0.047	0.211	0	1
Poor performers	0.052	0.223	0	1

Note: Statistics for 2495 observations for models in table 1 except for normative expectations with 2499 observations in these models (values of all other variable in these models were very close to those of the models with 2495 observations).

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